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Detour Steer — Summary

Decision algorithm for a robot that steers one way only: drive on, steer less, come back

1. Problem

The robot `mor_luam` has **one motor**. Turning it one way *steers* the wheel (counter-clockwise only). Turning it the other way *drives* forward. So the robot must **stop to steer** and can never steer while it drives. The old method steers straight at the goal, then drives. A goal a little to the **right** (φ close to 360°) then costs almost a full turn of the wheel.

2. Idea

Two straight legs whose headings differ by more than 180° can reach any direction. So instead of steering φ, the robot may:

1. drive a metres straight on (no steering),
2. steer only β^* $(180^\circ < \beta^* < \varphi)$,
3. drive b metres to the goal.

The best angle has a closed form (one number k decides everything):

$$k = \frac{d\,\omega}{v}\,|\sin\varphi|, \qquad \beta^* = 360^\circ - \arccos(k-1), \qquad a = \frac{d\,\sin(\beta^* - \varphi)}{\sin\beta^*}, \qquad b = \frac{d\,\sin\varphi}{\sin\beta^*}$$

d distance to the goal, ω steering speed, v driving speed, t_s settle time, t_b stop time between the two legs.

3. Decision algorithm

Procedure : *PlanLeg(φ, d)* // φ: counter-clockwise angle to the goal, d: distance
- $T_D \leftarrow \varphi/\omega + t_s + d/v$; // old method: steer φ, drive d
- **if** $\varphi \le 180^\circ$ **then return** *DIRECT*; // T1: steering straight is best
- $k \leftarrow (d\,\omega/v)\,|\sin\varphi|$;
- **if** $k \ge 2$ **then return** *DIRECT*; // T3: no detour can be faster
- $\beta^* \leftarrow 360^\circ - \arccos(k-1)$; compute a, b;
- $T_R \leftarrow a/v + t_b + \beta^*/\omega + t_s + b/v$;
- **if** $\beta^* < \varphi$ ***and*** $T_R < T_D$ **then return** *DETOUR(a, β^*, b)*; // detour only if really faster
- **return** *DIRECT*;

end

The robot runs this **after every move** (closed loop) with the real wheel angle (AS5600) and body heading (IMU). It steers to stop about 2° *short* on purpose: short costs a small extra turn, past costs almost a full turn. While driving it stops early and re-aims with a small turn ($\approx 18^\circ$).

4. Why it is correct and fast

- **T1** $\varphi \le 180^\circ$: steering straight is optimal.
- **T2** $\varphi > 180^\circ$: a best path has at most 2 legs.
- **T3** the time has exactly one minimum, at $\cos\beta^* = k - 1$.
- **T4** planning again after leg 1 keeps the same plan, so the closed loop is stable.
- **O(1)** per decision: $0.54\,\mu$s in Python, $944\times$ faster than a 3600-point search. T1–T4 are also checked by brute force in `detour_steer.py`: all PASS.

5. Results

Simulation (5000 random goals): Detour Steer is never slower than the old method. It saves 0.4 % when the wheel stops exactly, 2.3–6.9 % with random stop errors, and 12.8–18.3 % when the wheel tends to stop past its angle.

On the robot (goal 0.30 m away at $\varphi = 354°$, $v = 0.03$ m/s, $\omega = 35°/\text{s}$, 3 rounds):

round	old method (s)	Detour Steer (s)	saved (s)
1	24.17	20.83	3.34
2	22.94	20.77	2.17
3	23.36	22.33	1.03
mean	23.49	21.31	2.18 (9.3 %)

Detour Steer won 3 of 3 rounds. It steered 3.8 s less and drove 1.2 s more. All 6 runs ended within 2.5 mm of the goal.

Why it drifts right (both methods): one motor, so no course correction while driving; the drive torque turns the wheel module and gears a little (the AS5600 angle keeps changing and the front turns while it drives); the wheel slips, the floor is not flat and the weight is not balanced; and the wheel stops about 2° short on purpose, which points it a little right. Both methods end 1–3 cm right of their line, and an early re-aim (about 18°) brings them back.

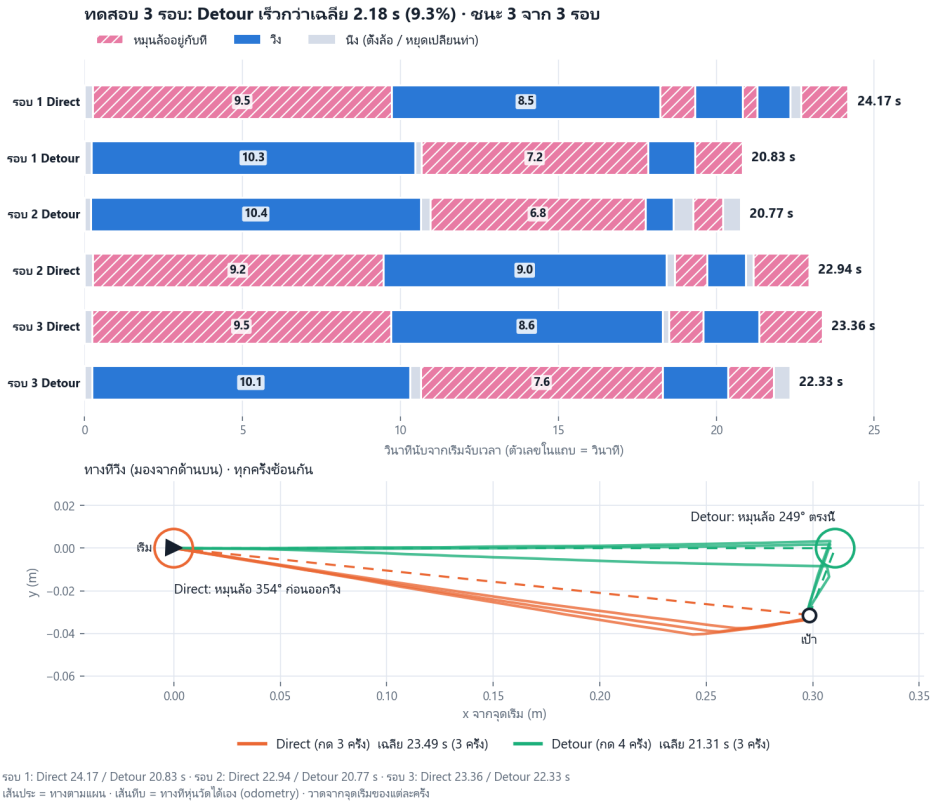


Figure 1. The 3 rounds, from the robot's own record (top: time line, bottom: measured paths).

6. Conclusion

One closed-form rule decides between "steer straight" and "drive on, steer less, come back". It costs $O(1)$ per decision, is never slower in simulation, and was 9.3 % faster on the real robot. No earlier work found gives this closed-form detour for a robot that must stop to steer one way only ("not found" is not proof that nobody did it).

Code, Git and video

All code (Python planner, simulators, robot firmware) and the full report:

`https://github.com/Mannaja78520/mor_luam`

- Planner and simulators: `algorithm_sim/` (runs without Docker: `pip install matplotlib`).
- Robot version: `firmware/src/algorithm/DetourSteer.cpp`.
- Full report: `algorithm_sim/docs/660610840_DetourSteer.pdf` (Thai), `report.pdf` (English).
- Install guide: `algorithm_sim/README.md`, `docs/INSTALL_TH.md` (Docker is the recommended install).

Video of the robot working:

`https://drive.google.com/drive/folders/11qOM6UoeH2c4nEuM0a0JO6eXTnr0eDud`

The clip shows the robot working in synchronize mode. There is no clip of the Detour Steer vs old method (Direct) comparison; that comparison comes from the robot's own record (Figure 1 and the results table).